Brendan J. Clark, Ph.D.

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EDUCATION

Ph.D. Atmospheric Science January 2025
Rutgers University, New Brunswick
Advisor: Prof. Alan Robock
Dissertation: *"Stratospheric Aerosol Climate Intervention Implications for Global Agriculture"*

M.S. Atmospheric Science November 2022
Rutgers University, New Brunswick
Advisor: Prof. Alan Robock
Thesis: *"Optimal climate intervention scenarios for crop production vary by nation"*

B.S. Environmental Science May 2020
University of Massachusetts, Amherst
Certificate in GIS, minor in Natural Resource Conservation

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate March 2025-present
Cornell University

- Leveraging advanced statistical, machine learning, and process-based modeling to assess future impacts to tree growth and agriculture
- Collaborating with CarbonPlan doing statistical downscaling and cloud storage of climate model simulations

Graduate Research Assistant August 2020-January 2025
Rutgers University

- Used the Community Earth System Model to assess impacts to global food production and nutritional quality under climate change and geoengineering
- Presented 15 first authors talks and posters
- Led cross-disciplinary research teams of up to 8 collaborators

PUBLICATIONS

PEER-REVIEWED ARTICLES

7. **B. Clark**, D. Visioni. Canopy structure Amplifies the Role of Radiation in Vegetation Responses of Stratospheric Aerosol Injection. **(in prep)**.
6. **B. Clark**, D. Visioni, S. Kothari, M. Lerdau. Land Models Likely Underestimate the Impact of Future Atmospheric Dryness on European Tree Growth. *Geophysical Research Letters* 53, e2026GL122759 **(2026)**. 10.1029/2026GL122759
5. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, G. Hoogenboom, J. Jaegermeyr. Maize yield changes under sulfate aerosol climate intervention using three global

gridded crop models. *Earth's Future* 13, e2024EF005269 **(2025)**. 10.1029/2024EF005269

4. N. Grant, A. Robock, L. Xia, J. Singh, **B. Clark**. Impacts on Indian agriculture due to stratospheric aerosol intervention using agroclimatic indices. *Earth's Future* 13, e2024EF005262 **(2025)**. 10.1029/2024EF005262
3. **B. Clark**, A. Robock, L. Xia, S. S. Rabin, J. R. Guarin, J. Jaegermeyr. Stratospheric aerosol climate intervention could reduce crop nutritional value. *Environmental Research Letters*, 20, 114083 **(2025)**. 10.1088/1748-9326/ae1151
2. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. Optimal climate intervention scenarios for crop production vary by nation. *Nature Food* 4, 902–911 **(2023)**. 10.1038/s43016-023-00853-3
1. F. Bowlick, **B. Clark**, et al. Understanding geocomputation education: A survey and syllabi informed review. Research in Geographic Education 23, 20-51 **(2022)**.

FUNDING AND AWARDS

- CESM Workshop Travel Award ($3,000) 2026
- Climate Intervention Network Travel Award ($1,000) 2025
- NSF NCAR Exploratory Allocation Award (500,000 CPU hours) 2025
- GeoMIP Annual Meeting Travel Funding Award ($3,500) 2025
- Rutgers Atmospheric Science Student Travel Support Award ($1,500) 2024
- Rutgers Atmospheric Science Student Travel Support Award ($1,200) 2023
- Rutgers Climate Institute Student Travel Support Award ($1,000) 2022
- NSF NCAR University Small Allocation Award (200,000 CPU hours) 2022
- Governor's Award in Recognition of Environmental Stewardship 2019
- John and Abigail Adams Scholarship Award ($7,000) 2016
- Stanley Z. Koplik Certificate of Mastery with Distinction Award 2016

TEACHING EXPERIENCE

Cornell University Spring 2026
Guest lecture, EAS 5555: Theory and Practice of Earth System Modeling

Rutgers University Spring 2026
Guest lecture, 01:090:297: Climate Intervention: Where Science Meets Ethics

Rutgers University Fall 2024
Guest lecture, 16:107:544: Modeling Climate Change

University of Massachusetts Fall 2019
Teaching Assistant, NRC 585: Introduction to GIS

Algonquin Regional High School 2018-2019
Substitute teacher, occasionally gave scientific lectures to students

CONFERENCE PRESENTATIONS (FIRST AUTHOR)

21. **B. Clark**. **Invited oral presentation**, June 2026. “Land models likely underestimate benefits to natural vegetation under stratospheric aerosol injection.” Gordon Research Conference, Newry, ME
20. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Oral presentation**, June 2026. “Land models likely underestimate the effect of atmospheric dryness on tree growth.” CESM Working Group Meeting, Boulder, CO
19. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Oral presentation**, March 2026. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” Coupled Model Intercomparison Project Community Workshop, Kyoto, Japan
18. **B Clark**, D. Visioni. **Poster**, March 2026. “Altered sunlight under stratospheric aerosol injection would have heterogenous impacts on vegetation.” Geoengineering Modeling Intercomparison Project annual meeting, Tokyo, Japan
17. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Oral presentation**, December 2025. “Land models likely underestimate the effect of atmospheric dryness on European tree growth.” American Geophysical Union meeting, New Orleans, LA
16. **B Clark**, D. Visioni, S. Kothari, M. Lerdau. **Poster**, December 2025. “Using Dendrometer Data to Inform Projections of Carbon Storage under Solar adiation Modification.” American geophysical Union meeting, New Orleans, LA
15. **B. Clark. Poster**, May 2025 “Statistcally downscaled and bias adjusted ARISE-SAI and SSP2-4.5 scenarios.” DEGREES Global Forum, Cape Town, South Africa
14. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, December 2024. “Stratospheric Aerosol Climate Intervention Could Reduce the Nutritional Value of Maize and Rice.” American Geophysical Union meeting, Washington D.C.
13. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, July 2024. “Sulfate Aerosol Climate Intervention Impacts on Maize Yield and Protein in Three Global Gridded Crop Models.” Geoengineering Modeling Intercomparison Project annual meeting, Cornell University
12. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral presentation**, February 2024. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” Gordon Research Seminar, Newry, ME
11. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Poster**, February 2024. “Stratospheric Aerosol Climate Intervention Impacts on Crop Protein Content.” Gordon Research Conference, Newry, ME
10. **B. Clark,** A. Robock, L. Xia, S. Rabin, J Jaegermeyr, J. Guarin. **Oral Presentation**, December 2023. “Stratospheric Aerosol Climate Intervention Could Negatively Impact Crop Nutritional Quality.” American Geophysical Union meeting, San Francisco, CA
9. **B. Clark**, L. Xia, A. Robock, S. Tilmes, J. H. Richter, D. Visioni, S. S. Rabin. **Invited oral presentation**, July 2023. “The Optimal Climate Intervention Scenario for Crop Production Varies by Nation.” Exteter, UK

8. **B. Clark,** A. Robock, L. Xia, S. Rabin, J. Singh, N. Grant. **Poster**, July 2023. "Rutgers Impact Studies of Climate Intervention (RISCI) Laboratory Group Overview." Geoengineering Modeling Intercomparison Project annual meeting, Exeter, UK
7. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, June 2023. "A Proposal for a Multi-Crop Model Assessment of Stratospheric Aerosol Climate Intervention." AgMIP-GGCMI meeting, Columbia University
6. **B. Clark,** A. Robock, L. Xia. **Poster**, December 2022. "Can Crop Production be used as a Metric to Design Climate Intervention?" American Geophysical Union meeting, Chicago, IL
1. **B. Clark,** A. Robock, L. Xia. **Poster**, June 2022. "Impacts on Crop Production from Stratospheric Aerosol Climate Intervention: A Multi-Scenario Overview." Gordon Research Conference, Newry, ME
4. **B. Clark,** A. Robock, L. Xia. **Invited oral presentation**, May 2022. "Crop Impacts from Stratospheric Aerosol Injection: A Multi-Scenario Overview." University of Potsdam, Germany
1. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, February 2022. "Discrepancies Between Fully Coupled and Offline CLM5 Crop Simulations." NCAR Land Modeling Working Group Meeting, Boulder, CO
2. **B. Clark,** A. Robock, L. Xia. Poster, December 2021. "Impacts on Crop Production from Stratospheric Aerosol Injection" American Geophysical Union, New Orleans, LA
1. **B. Clark,** A. Robock, L. Xia. **Oral presentation**, October 2021. "Depicting Information and Remembering Your Audience: Impacts on Crop Production from Stratospheric Aerosol Climate Intervention" Climate Engineering in Context Conference, University of Potsdam, Germany

TECHNICAL SKILLS

- Python
  - Data analysis packages including xarray, numpy, pandas, scipy, and dask
  - Data visualization packages including matplotlib, seaborn, and plotly
  - Statistical modeling and machine learning packages including SciPy, statsmodels, scikit-learn, and seaborn
  - Jupyter notebooks and Google Colab
- Cloud computing, including AWS
- HTML, Linux, JavaScript, MATLAB, Fortan, R
- Version control and collaborative work using Git and GitHub
- Shell and command line – bash shell, batch scripting
- Parallelization using Python multiprocessing and Slurm scheduling
- High-performance computing environments including university clusters and NCAR's Casper and Derecho

SERVICE

- Journal refereeing for: *Earth's Future, Atmospheric Chemistry and Physics, Journal of Geophysical Research - Atmospheres, Environmental Research -Climate, PLOS Climate*
- The Degrees Initiative grant proposal reviewer
- Reflective grant proposal reviewer
- Physicists Coalition for Nuclear Threat Reduction
- Science Policy and Advocacy at Rutgers

PROFESSIONAL AFFILIATIONS

- American Geophysical Union
- Ecological Society of America
- Agricultural Modeling Intercomparison Project
- Global Gridded Crop Modeling Intercomparison Project
- Geoengineering Modeling Intercomparison Project
- Climate Intervention Biology Working Group
- Inter-Sectoral Impact Model Intercomparison Project

SELECTED MEDIA COVERAGE

- "Climate Intervention Technologies May Create Winners and Losers in World Food Supply," *Rutgers Today*
- "Blocking the sun may fight climate change but kill our crops," *The Daily Beast*
- "Climate Tactics May Lower Crop Nutritional Value," *Mirage News*
- "Getting warmer: Slower forest growth means less carbon storage," *Cornell Chronicle*
- "What happens to our food supply when we dim the sun?" *Climate Chat Podcast*